Crowbar Thyristor for the 3 kV / 3600 A Static-Switch Test Bench

Device Selection and Passive Self-Triggering Gate Circuit

Rev. 00 — 2026-09-09

Rev.	Date	Description
00	2026-09-09	First issue. Device selection, stress analysis, passive gate circuit.

Abstract

A thyristor crowbar with a series dump inductor and resistor extracts the energy of a 2 mH, 3600 A inductive load when a 3 kV IGBT static switch opens. This note quantifies the electrical stress on the crowbar thyristor, justifies the choice of a 6.5 kV symmetric phase-control thyristor (Hitachi Energy 5STP 26N6500), and explains why the intrinsic self-firing mechanisms of a thyristor (critical $\mathrm{d}v/\mathrm{d}t$ and forward breakover) must not be used as they are, but can be turned into a reliable, fully passive trigger by a floating anode-to-gate network. Design equations and starting values for that network are given, together with the coordination rules with the MOV clamp of the main switch and the checks to be made in simulation.

1 Scope and circuit under study

The circuit is the static-switch test bench of Fig. 1: a DC source $V_s = 3000\,\mathrm{V}$ feeds an RL load ($R = 0.8\,\Omega$, $L = 2\,\mathrm{mH}$) through an IGBT breaker S (four 5SNA 1000G650300 in parallel). The breaker is opened under load at $I_0 = 3600\,\mathrm{A}$; a varistor bank across the breaker clamps its voltage at $V_{\mathrm{MOV}} \approx 5.5\,\mathrm{kV}$. The load energy is extracted by the crowbar branch in parallel with the load: thyristor T, dump inductor $L_c = 12.5\,\mu\mathrm{H}$ and dump resistor $R_D = 0.42\,\Omega$, with an RC snubber $C_{sn} = 2\,\mu\mathrm{F}$, $R_{sn} = 64\,\Omega$ across the branch. The bench must perform of the order of 10^6 opening cycles.

Two facts shape the whole analysis:

1. In service the crowbar thyristor is *reverse* biased at $-V_s = -3\,\mathrm{kV}$. Forward voltage appears only when S opens and the load inductance reverses the load voltage, which is exactly when the crowbar must fire.

2. The forward voltage available to the crowbar is bounded by the MOV clamp of the breaker:

$$v_{AK,\max} = V_{\mathrm{MOV}} - V_s \approx 5.5\,\mathrm{kV} - 3.0\,\mathrm{kV} = 2.5\,\mathrm{kV}. \tag{1}$$

   Any firing threshold of the crowbar must lie well below this value, otherwise the MOV takes the load current and the energy.

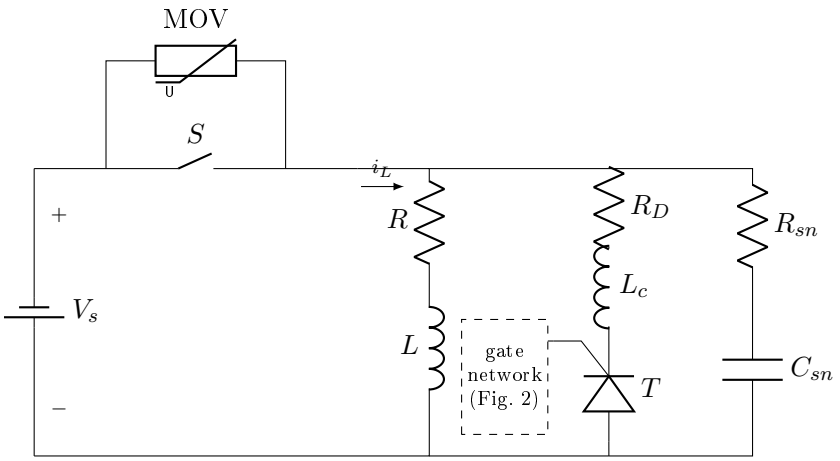


Figure 1: Static-switch test bench with dump-resistor crowbar. In service (S closed) the load voltage is $+V_s$ and the thyristor blocks in the *reverse* direction; forward voltage appears across T only while S opens.

2 Electrical stress on the crowbar thyristor

2.1 Conduction: charge, $\int i^2\,\mathrm{d}t$, energy

Once the load current has commutated into the crowbar branch, it decays through $R + R_D$ with

$$\tau = \frac{L}{R + R_D} = \frac{2\,\mathrm{mH}}{1.22\,\Omega} \approx 1.64\,\mathrm{ms}. \tag{2}$$

The conduction quantities follow directly:

$$Q = I_0\,\tau \approx 5.9\,\mathrm{C}, \tag{3}$$

$$\int i^2\,\mathrm{d}t = \frac{I_0^2\,\tau}{2} \approx 10.6 \times 10^3\,\mathrm{A^2s}, \tag{4}$$

$$E_L = \tfrac{1}{2} L I_0^2 \approx 13.0\,\mathrm{kJ}, \qquad E_{R_D} = E_L\,\frac{R_D}{R + R_D} \approx 4.5\,\mathrm{kJ}, \qquad E_R \approx 8.5\,\mathrm{kJ}. \tag{5}$$

The thyristor conduction energy, with the on-state model $v_T = V_{T0} + r_T\,i$ of the 5STP 26N6500 ($V_{T0} = 1.12\,\mathrm{V}$, $r_T = 0.29\,\mathrm{m\Omega}$), is

$$E_T = V_{T0}\,Q + r_T \int i^2\,\mathrm{d}t \approx 6.6 + 3.1 \approx 10\,\mathrm{J}. \tag{6}$$

Spread over the silicon of a 100 mm wafer (thermal capacity of the order of 10 J/K), this gives $\Delta T_j \lesssim 1\,\mathrm{K}$ per event. The crowbar thyristor is therefore *not* sized by current or by $\int i^2\mathrm{d}t$; it is sized by blocking voltage, turn-on $\mathrm{d}i/\mathrm{d}t$ and cycle count. The current reaches the holding level ($I_H \approx 1\,\mathrm{A}$) after $\tau \ln(I_0/I_H) \approx 13\,\mathrm{ms}$ and the device turns off naturally, without reverse recovery.

2.2 Commutation: $\mathrm{d}i/\mathrm{d}t$ and MOV energy

While S opens, the breaker voltage is held at V_{MOV} by the varistor bank; the forward voltage (1) drives the load current into L_c:

$$\frac{\mathrm{d}i}{\mathrm{d}t} = \frac{V_{\mathrm{MOV}} - V_s}{L_c} = \frac{2.5\,\mathrm{kV}}{12.5\,\mu\mathrm{H}} = 200\,\mathrm{A/\mu s}, \tag{7}$$

$$t_{\mathrm{comm}} = \frac{L_c\,I_0}{V_{\mathrm{MOV}} - V_s} = 18\,\mu\mathrm{s}, \tag{8}$$

$$E_{\mathrm{MOV,comm}} \approx \tfrac{1}{2}\,V_{\mathrm{MOV}}\,I_0\,t_{\mathrm{comm}} \approx 180\,\mathrm{J}. \tag{9}$$

The MOV bank thus sees about 180 J per opening instead of the ∼16 kJ it would absorb without the crowbar, which is what makes the 10^6-cycle target credible. The turn-on $\mathrm{d}i/\mathrm{d}t$ is set by the MOV clamp level and L_c, not by the triggering method: it is the same whether the thyristor fires at 50 V or at 2 kV. It must be compared with the *repetitive* critical $\mathrm{d}i/\mathrm{d}t$ of the datasheet (for large phase-control thyristors of the order of a few hundred A/µs); if the margin is insufficient, increase L_c or lower the MOV clamp.

2.3 $\mathrm{d}v/\mathrm{d}t$ at breaker opening

At the instant S turns off, the load current initially flows into the capacitances hanging on the load node; with the branch snubber alone as an upper bound,

$$\left.\frac{\mathrm{d}v}{\mathrm{d}t}\right|_{\max} \approx \frac{I_0}{C_{sn}} = \frac{3600\,\mathrm{A}}{2\,\mu\mathrm{F}} = 1.8\,\mathrm{kV/\mu s}, \tag{10}$$

so the thyristor voltage swings from −3 kV through zero into forward bias in about 2 µs. The breaker's own RCD snubbers in parallel reduce this figure, but it remains of the same order as the critical $\mathrm{d}v/\mathrm{d}t$ of a phase-control thyristor (typically 1000–2000 V/µs). This is the key observation of Section 4: left to itself, the device is likely to self-fire at breaker opening, in an uncontrolled way.

3 Device selection

3.1 Requirements

Table 1: Crowbar thyristor requirements and 5STP 26N6500 data.

Quantity	Requirement	5STP 26N6500	Note
Reverse blocking, continuous DC	3.0 kV	$V_{RRM} = 6.5$ kV	cosmic-ray FIT, see text
Forward blocking, transient	≤ 2.5 kV	$V_{DRM} = 6.5$ kV	bounded by MOV
Peak current	3600 A	$I_{TSM} = 65$ kA	
$\int i^2\mathrm{d}t$ per event	10.6×10^3 A^{2}s	$\sim 2 \times 10^7$ A^{2}s	from I_{TSM}, 10 ms
Turn-on $\mathrm{d}i/\mathrm{d}t$, repetitive	200 A/µs	datasheet	check margin
$\mathrm{d}v/\mathrm{d}t$ at turn-on	up to 1.8 kV/µs	1000–2000 V/µs	gate network required
Cycles	$\sim 10^6$	—	$\Delta T_j \lesssim 1$ K
Gate trigger current	—	$I_{GT} \leq 0.4$ A	

3.2 Voltage class: why 6.5 kV

The device blocks 3 kV DC continuously in the reverse direction for the whole life of the bench. For high-voltage silicon devices the dominant random failure mechanism under continuous DC blocking is cosmic-ray induced breakdown, whose rate rises very steeply with the ratio of applied to rated voltage. A 4.5 kV device would be operated at about 67 % of its rating, a 6.5 kV device at 46 %. The FIT-versus-V_{DC} curves of the manufacturer must be consulted for the chosen part, but the 6.5 kV class is the natural choice; it also gives comfortable margin against the transients at breaker closing, when the device sees a 0→−3 kV step.

The 5STP 26N6500 (Hitachi Energy, formerly ABB) is a symmetric 6.5 kV phase-control thyristor with interdigitated amplifying gate, $I_{T(RMS)} = 4410$ A, $I_{TSM} = 65$ kA [1]. A Simscape model of this device with Lauritzen–Ma reverse recovery already exists, so the simulation work carries over. Its current margins in this application are very large (three orders of magnitude on $\int i^2\mathrm{d}t$); a smaller wafer of the same 6.5 kV family would be adequate thermally and would

reduce clamping force and cost. The 100 mm device is retained here for the $\mathrm{d}i/\mathrm{d}t$ margin and because it is already modelled.

3.3 Why not an asymmetric pulse-power thyristor

Pulse-power thyristors (e.g. the Dynex range based on GTO structures, 3300 V and 4500 V, highly interdigitated gate, very high $\mathrm{d}i/\mathrm{d}t$) are marketed also for crowbar duty [5]. They do not fit here for two reasons: they are asymmetric and do not block the −3 kV seen in service (a series diode would be needed), and the $\mathrm{d}i/\mathrm{d}t$ of 200 A/µs imposed by L_c is well within the reach of a phase-control device. A symmetric PCT is the simpler and more robust choice.

4 Self-firing mechanisms and why they must not be used as they are

4.1 $\mathrm{d}v/\mathrm{d}t$ firing

A rising forward voltage drives a displacement current $C_{J2}\,\mathrm{d}v/\mathrm{d}t$ through the blocking junction J_2. This current flows laterally in the p-base towards the cathode shorts; where the lateral voltage drop across the p-base under the n^+ emitter reaches ≈ 0.7 V, the emitter injects and the thyristor turns on [2]. The location is wherever the p-base resistance is highest or the shorts are farthest, i.e. a random point of the cathode area, not the gate region. The datasheet "critical $\mathrm{d}v/\mathrm{d}t$" is the ramp the device is guaranteed *not* to fire on at $T_{j,\max}$; the actual firing ramp is unspecified, may be several times higher, and falls with temperature because leakage adds to the displacement current.

4.2 Forward breakover

When the forward voltage exceeds V_{BO} the avalanche multiplication at J_2 satisfies the regenerative condition $\alpha_{npn}\,\alpha_{pnp}\,M \to 1$ and the device turns on [2]. V_{BO} is not a datasheet parameter: it lies at roughly 1.1–1.3×V_{DRM}, varies with temperature and from device to device, and the turn-on initiates at the point of maximum field (junction termination or gate periphery). The undesired ignition of a thyristor occurs at a random, localized spot of the cathode junction and may destroy the device through the concentrated turn-on losses [3].

4.3 Consequences in this circuit

- In both mechanisms the initially conducting area is small and unrelated to the amplifying gate; the $\mathrm{d}i/\mathrm{d}t$ that follows (200 A/µs here) must then be carried by that small area. The datasheet $\mathrm{d}i/\mathrm{d}t$ rating does not apply. Over 10^6 cycles this is not an acceptable operating mode.
- Static self-firing on V_{BO} is not even reachable: the forward voltage is limited to 2.5 kV by the MOV (1), far below the $V_{BO} > 6.5$ kV of the chosen device. The MOV would absorb the full 16 kJ and the crowbar would never fire.
- Dynamic self-firing is, on the contrary, *likely* to happen: (10) is at the critical $\mathrm{d}v/\mathrm{d}t$ level. Without a gate network the device will fire, at an unpredictable point of the wafer, at an unpredictable forward voltage.

The useful conclusion is not to suppress these mechanisms but to *pre-empt* them: the same anode voltage that would cause an uncontrolled ignition is used to inject a proper gate current, so that turn-on always starts at the amplifying gate. This is the "protective firing" concept of HVDC valves, where a break-over diode (BOD) between anode and gate converts an overvoltage

into a regular gate pulse [3, 4]. Here the roles are extended: the dynamic path is the primary trigger, the static path is the back-up.

5 Passive self-triggering gate network

5.1 Topology

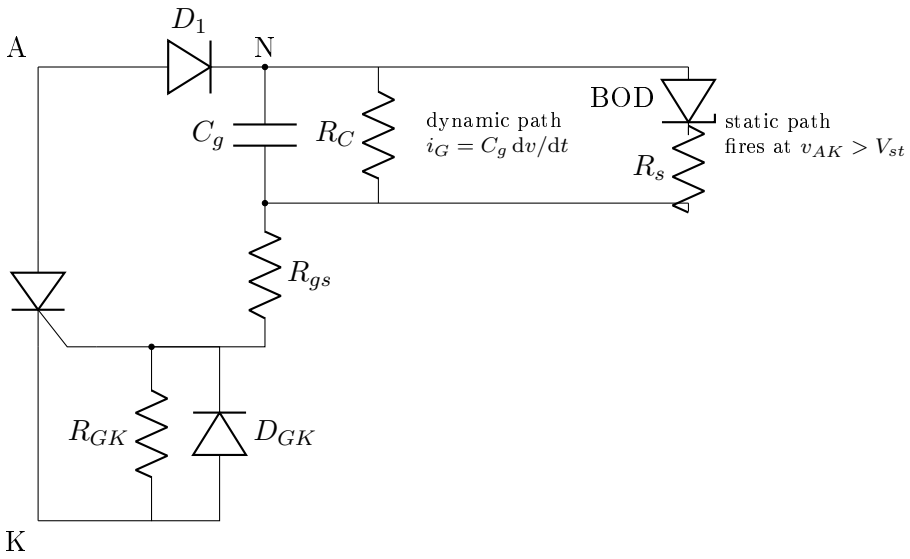


Figure 2: Floating anode-to-gate trigger network. D_1 isolates the network during reverse blocking; C_g converts the forward $\mathrm{d}v/\mathrm{d}t$ into gate current; the BOD (or TVS stack) with R_s fires the gate if the forward voltage rises slowly beyond V_{st}; R_C resets C_g; R_{GK} and D_{GK} protect the gate.

All components are referenced to the cathode and float with it: no auxiliary supply, no fibre-optic link, no isolation transformer. The network works as follows.

D_1 – **reverse isolation.** With $v_{AK} < 0$ (all of service life) D_1 is reverse biased and node N sits at cathode potential: no current reaches the gate. This is essential, because a positive gate current while the thyristor blocks reverse voltage forward-biases J_3 and J_2 and is amplified by the pnp transistor gain into an increased reverse leakage concentrated near the gate; manufacturers exclude this condition. D_1 also keeps the static-path elements from conducting in their forward direction at $-3\,\mathrm{kV}$. Its blocking voltage must cover V_s plus closing transients with margin: a string rated $\geq 6\,\mathrm{kV}$ (e.g. 3–4 fast diodes of 1.6–2.2 kV with sharing resistors), or a HV diode stack. Its current is only the gate current, tens of amperes for microseconds.

Dynamic path – C_g. As soon as v_{AK} crosses zero and rises, D_1 conducts and

$$i_G \approx C_g \frac{\mathrm{d}v_{AK}}{\mathrm{d}t} \tag{11}$$

flows into the gate through R_{gs} (valid for $R_{gs}C_g$ much shorter than the rise time). Firing occurs when the injected current exceeds the gate trigger current plus what R_{GK} diverts:

$$\left.\frac{\mathrm{d}v}{\mathrm{d}t}\right|_{th} = \frac{I_{GT} + V_{GT}/R_{GK}}{C_g}. \tag{12}$$

The thyristor therefore turns on at the amplifying gate within a few tens of volts of forward bias, long before the intrinsic $\mathrm{d}v/\mathrm{d}t$ mechanism could act.

R_C – reset. C_g charges to the forward voltage present at firing (up to 2.5 kV); after turn-on D_1 blocks and C_g discharges through R_C with $\tau_C = R_C C_g$, which must be much longer than the voltage rise (µs) and much shorter than the cycle period. R_C dissipates $\frac{1}{2} C_g v_{AK}^2$ per event.

Static path – BOD or TVS stack with R_s. If the forward voltage rises too slowly for the dynamic path (abnormal breaker opening, fault), the BOD conducts once $v_{AK} > V_{st}$ and drives the gate through R_s, which limits the current to $(v_{AK} - V_{st})/R_s$. Since no forward voltage exists in service, V_{st} can be low; it must only exceed ringing and noise and remain far below (1). Genuine BODs are available from HVDC gate-unit suppliers; at a threshold of a few hundred volts a stack of high-energy TVS diodes (5KP/15KP class) does the same job.

R_{GK}, D_{GK}. R_{GK} gives noise immunity and sets the threshold (12); D_{GK} (antiparallel) limits negative gate voltage to well below V_{RGM}, e.g. against the small displacement current through the D_1 junction capacitance at breaker closing.

5.2 Starting values

Table 2: Starting values for the gate network (5STP 26N6500, $I_{GT} \leq 0.4\,\text{A}$, $V_{GT} \approx 3\,\text{V}$ to be confirmed on the datasheet).

Element	Value	Design check
C_g	4.7 nF, ≥ 4 kV DC	$i_G = 2.4\,\text{A}$ at 500 V/µs, 8.5 A at 1.8 kV/µs; compare peak with I_{FGM}
R_{GK}	10 Ω	threshold (12): (0.4 + 0.3) A/4.7 nF ≈ 150 V/µs
R_{gs}	10–15 Ω	$R_{gs}C_g \approx 70\,\text{ns} \ll$ rise time; damps the C_g–lead-inductance loop, $2\sqrt{L_{lead}/C_g} \approx 13\,\Omega$ for $L_{lead} = 0.2\,\mu\text{H}$
R_C	22 kΩ, HV	$\tau_C \approx 100\,\mu\text{s}$; $E = \frac{1}{2} C_g (2.5\,\text{kV})^2 \approx 15\,\text{mJ}$ per event
BOD / TVS	$V_{st} \approx 400 - -500\,\text{V}$	e.g. 3 × 150–180 V TVS 5KP/15KP; $V_{st} \ll 2.5\,\text{kV}$
R_s	200 Ω, pulse rated	$i_{G,\max} = (2500 - 500)/200 = 10\,\text{A}$; ∼15 mJ in the stack per event (worst case)
D_1	string ≥ 6 kV, fast	blocks $-V_s$ in service; sharing R/C across the string
D_{GK}	fast or Schottky, ≥ 15 A pulse	clamps $v_{GK} < 0$

5.3 Optional active trigger

If the breaker controller can issue a gate command, it should be used as the primary trigger (pulse transformer or fibre-optic gate unit) with the passive network of Fig. 2 left in place as an independent back-up – the same philosophy as HVDC valves. The command must be issued together with, not before, the IGBT turn-off: a positive gate pulse applied while the thyristor still blocks −3 kV produces the reverse-leakage amplification described above.

6 Coordination with the MOV clamp

1. **Firing level:** $V_{st} \ll V_{\text{MOV}} - V_s$. With $V_{st} \approx 0.5\,\text{kV}$ and 2.5 kV available the margin is a factor five; the dynamic path fires even earlier.

2. $\mathrm{d}i/\mathrm{d}t$**:** $(V_{\mathrm{MOV}} - V_s)/L_c = 200\,\mathrm{A/\mu s}$, independent of the trigger. To be checked against the repetitive datasheet value; the two knobs are L_c and the MOV clamp level.

3. **MOV duty:** about 180 J per opening during the 18 µs commutation interval; the varistor bank becomes a back-up clamp with negligible ageing.

4. **Branch snubber:** at firing C_{sn} discharges through R_{sn} into the thyristor, $v_{AK}/R_{sn} \leq 40\,\mathrm{A}$ and $\frac{1}{2}C_{sn}v_{AK}^2 \leq 6\,\mathrm{J}$ in R_{sn}: harmless.

7 Verification in simulation

With the existing Simscape model of the 5STP 26N6500 and the gate network added:

- Plot v_{AK} (device terminals, not branch voltage), i_G and i_A at breaker opening. Confirm that i_G exceeds I_{GT} while v_{AK} is still low, and record the forward voltage at which anode current starts.
- Extract $\mathrm{d}i_A/\mathrm{d}t$ during the first 20 µs and compare with the repetitive rating.
- Sweep the effective capacitance at the load node (branch snubber plus breaker snubbers) to cover the range of $\mathrm{d}v/\mathrm{d}t$ from 200 V/µs to 2 kV/µs; the static path should take over below the threshold (12).
- Breaker closing: v_{AK} steps from 0 to $-3\,\mathrm{kV}$; check that the gate stays quiet (v_{GK} within $\pm$ a few volts) and that D_1 sees no overvoltage.
- Abnormal case: breaker re-closing while the crowbar still conducts. The source then forces the crowbar current to zero through L_c at $V_s/L_c = 240\,\mathrm{A/\mu s}$ with reverse recovery; check the reverse voltage overshoot across T and the branch snubber. The turn-off time t_q is irrelevant in normal operation, since forward voltage returns only at the next cycle.

8 Conclusions

The crowbar thyristor of the 3 kV / 3600 A bench dissipates about 10 J per event and is sized by voltage, $\mathrm{d}i/\mathrm{d}t$ and cycle count. A symmetric 6.5 kV phase-control thyristor such as the 5STP 26N6500 is the appropriate choice: it blocks the 3 kV reverse DC of service life with adequate cosmic-ray margin, and its $\mathrm{d}i/\mathrm{d}t$ capability matches the $200\,\mathrm{A/\mu s}$ imposed by L_c and the MOV clamp. The intrinsic self-firing mechanisms must not be relied upon: forward breakover is unreachable below the MOV clamp, and $\mathrm{d}v/\mathrm{d}t$ firing is likely but uncontrolled. A floating anode-to-gate network – blocking diode, coupling capacitor, BOD/TVS stack – turns both into a regular gate-triggered turn-on with no auxiliary supply, and can be combined with an active command as primary trigger. Firing level and $\mathrm{d}i/\mathrm{d}t$ are coordinated with the MOV clamp through $V_{st} \ll V_{\mathrm{MOV}} - V_s$ and $(V_{\mathrm{MOV}} - V_s)/L_c$.

References

[1] Hitachi Energy (formerly ABB Semiconductors), "Phase Control Thyristor 5STP 26N6500," data sheet.

[2] B. J. Baliga, *Fundamentals of Power Semiconductor Devices*. New York: Springer, 2008, ch. 8 (Thyristors).

[3] "Protective circuitry for a thyristor," U.S. Patent 4 651 251 (break-over diode protective firing, Nelson River HVDC precedent).

[4] J. Arrillaga, *High Voltage Direct Current Transmission*, 2nd ed. London: IEE, 1998 (thyristor valves, protective firing).

[5] Dynex Semiconductor, "Pulsed power thyristors," product information, accessed Sept. 2026.

[6] ABB Switzerland Ltd, Semiconductors, "Design of RC snubbers for phase control applications," Application Note 5SYA 2020.